A. HEURISTIC EVALUATION

#: 1	**Problem/Good:** Problem
Name: Overwhelming Number of Search Results	
Relevant heuristic: Aesthetic & Minimalist Design	
Evidence of issue: When users search for food (e.g., “cheap food near me”), Yelp displays a long list of restaurants with ratings, images, ads, and metadata all at once.	
Detailed explanation: The interface presents too much information simultaneously without prioritization. Users must scan through many options and mentally compare them. This increases cognitive load and makes it difficult to quickly identify the best choice. Instead of simplifying decision-making, the system shifts the burden entirely onto the user.	
Severity or Benefit (minor, major, critical): 3 – Major **Justification:** ● Frequency: Occurs every time users perform a search ● Impact: Leads to decision fatigue and slower task completion ● Persistence: Continues throughout browsing ● Weighting: Major because it directly affects core functionality (choosing food)	

#: 2	**Problem/Good:** Problem
Name: Lack of Personalized Recommendations	
Relevant heuristic: Flexibility & Efficiency of Use	
Evidence of issue: Search results are largely generic and based on ratings, distance, or ads rather than user-specific intent (e.g., budget, urgency, mood).	
Detailed explanation: The system does not adapt to individual user needs in real-time. Users must manually adjust filters to approximate their preferences, which is inefficient. A more intelligent system would interpret user intent and provide tailored suggestions.	
Severity or Benefit (minor, major, critical): 3 – Major	

Justification: ● Frequency: Occurs in all search scenarios ● Impact: Reduces efficiency and relevance of results ● Persistence: Persists across sessions ● Weighting: Major because it limits usability and personalization

#: 3	**Problem/Good:** Problem
Name: Inefficient Filtering System	
Relevant heuristic: User Control & Freedom	
Evidence of issue: Filters (price, distance, rating) must be manually selected and adjusted multiple times to refine results.	
Detailed explanation: The filtering system requires users to translate their needs into predefined categories. This creates friction, especially when users have vague or complex goals (e.g., “cheap but good dinner nearby right now”).	
Severity or Benefit (minor, major, critical): 3 – Major **Justification:** ● Frequency: Frequent during search refinement ● Impact: Slows down decision-making ● Persistence: Repeats with every search ● Weighting: Major due to inefficiency in core workflow	

#: 4	**Problem/Good:** Problem
Name: High Cognitive Load During Comparison	
Relevant heuristic: Recognition Rather Than Recall	
Evidence of issue: Users must remember and compare multiple restaurants while scrolling.	
Detailed explanation: There is no system support for comparing or narrowing choices. Users rely on memory, which increases mental effort and reduces efficiency.	
Severity or Benefit (minor, major, critical): 3 – Major	

Justification:

- Frequency: Occurs when browsing options
- Impact: Slows decision-making
- Persistence: Throughout browsing
- Weighting: Major due to cognitive burden

#: 5	**Problem/Good:** Problem
Name: Poor Visibility of Ranking Logic	
Relevant heuristic: Visibility of System Status	
Evidence of issue: it is unclear why certain restaurants appear at the top (e.g., ads vs relevance vs rating).	
Detailed explanation: Users cannot easily understand how results are prioritized. This reduces trust in the system and makes decision-making less reliable.	
Severity or Benefit (minor, major, critical): 2 – Minor **Justification:** ● Frequency: Occurs during all searches ● Impact: Causes confusion but not blocking ● Persistence: Continuous ● Weighting: Minor but affects trust	

#: 6	**Problem/Good:** Problem
Name: Lack of Context-Aware Recommendations	
Relevant heuristic: Match Between System & Real World	
Evidence of issue: The app does not consider context such as time, urgency, or user situation.	
Detailed explanation: Users think in terms like “quick lunch” or “late-night cheap food,” but Yelp requires manual filtering instead of understanding natural intent.	
Severity or Benefit (minor, major, critical): 3 – Major	

Justification: ● Frequency: Occurs in all real-world scenarios ● Impact: Misalignment with user thinking ● Persistence: Persistent ● Weighting: Major due to mismatch with real-world behavior

#: 7	**Problem/Good:** Good
Name: Clear Rating and Review System	
Relevant heuristic: Visibility of System Status	
Evidence of issue: Each restaurant displays ratings, number of reviews, and price level prominently in search results.	
Detailed explanation: The system provides immediate feedback about restaurant quality through ratings and review counts. This allows users to quickly assess general reliability without opening each listing. The visual representation (stars and numbers) reduces the effort required to interpret quality.	
Severity or Benefit (minor, major, critical): N/A **Justification:** ● Frequency: Present in all search results ● Impact: Helps users quickly evaluate options ● Persistence: Consistently useful ● Weighting: Major benefit because it supports fast decision-making	

#: 8	**Problem/Good:** Good
Name: Map Integration for Location Awareness	
Relevant heuristic: Match Between System & Real World	
Evidence of issue: Yelp integrates a map view showing restaurant locations relative to the user.	

Detailed explanation: The map aligns with real-world spatial understanding, allowing users to visualize distance and proximity. This reduces the need to interpret raw address data and supports intuitive navigation decisions.
Severity or Benefit (minor, major, critical): N/A **Justification:** ● Frequency: Frequently used when browsing nearby options ● Impact: Improves spatial understanding ● Persistence: Continues to provide value ● Weighting: Major benefit for location-based decisions

#: 9	**Problem/Good:** Good
Name: Rich Visual Content (Photos**)**	
Relevant heuristic: Recognition Rather Than Recall	
Evidence of issue: Restaurant listings include food images and interior photos.	
Detailed explanation: Visual content helps users quickly recognize appealing options without relying on textual descriptions. This reduces cognitive effort and improves engagement.	
Severity or Benefit (minor, major, critical): N/A **Justification:** ● Frequency: Present in most listings ● Impact: Enhances user engagement and understanding ● Persistence: Continues to aid decision-making ● Weighting: Moderate benefit	

B. HYPOTHESIS:

Attitude Hypotheses

1. We believe that users feel overwhelmed when browsing food apps because too many options are presented without clear prioritization. [Proven]
2. We believe that users feel frustrated when they cannot quickly decide what to eat due to lack of guidance or recommendations. [Proven]
3. We believe that users expect modern apps to provide personalized suggestions rather than requiring manual filtering. [Proven]

4. We believe that users prefer convenience and speed over exploring large numbers of options. [Unknown]

Behavior Hypotheses

1. We believe that users repeatedly scroll through multiple restaurant options before making a decision. [Proven]
2. We believe that users rely heavily on ratings and reviews when choosing restaurants. [Proven]
3. We believe that users frequently apply and adjust filters (price, distance, rating) to refine search results. [Proven]
4. We believe that users abandon the app or delay decisions when overwhelmed by too many choices. [Unknown]

Feature Hypotheses

1. We believe that an AI assistant that recommends the “best option” will reduce decision time. [Proven]
2. We believe that allowing users to input natural language queries (e.g., “cheap dinner nearby”) will improve usability. [Proven]
3. We believe that reducing the number of displayed options will improve user satisfaction. [Unknown]
4. We believe that proactive recommendations (instead of manual filtering) will improve efficiency and engagement. [Proven]

C. THEMATIC ANALYSIS:

Theme 1: Decision Fatigue and Overwhelming Choices
- Users frequently report difficulty choosing due to too many restaurant options
- Scrolling through long lists increases frustration
- Users spend more time deciding than ordering

Theme 2: Ineffective Recommendations
- Users feel recommendations are not personalized
- Suggested restaurants often don’t match preferences
- Ratings alone are not enough to guide decisions

Theme 3: Complex and Inefficient Filtering
- Users rely on filters but find them tedious
- Filters do not fully reflect real needs (e.g., mood, urgency)
- Repeated adjustments are common

Theme 4: High Cognitive Load
- Users must compare multiple restaurants manually
- Remembering options becomes difficult

- Information overload reduces usability

Theme 5: Trust and Transparency Issues

- Users question why certain restaurants are ranked higher
- Sponsored results create confusion
- Lack of transparency reduces trust

D. PROBLEM STATEMENT

1. A busy user needs a faster way to decide what to eat in order to avoid spending excessive time scrolling through restaurant options and reduce decision fatigue.
2. A food app user needs personalized recommendations based on their preferences (e.g., budget, distance, context) in order to find relevant options without manually adjusting multiple filters.
3. A time-sensitive user (e.g., lunch break) needs a quick "best option" suggestion in order to make efficient decisions without comparing multiple restaurants.
4. A new user needs a simplified and guided experience in order to navigate food options without feeling overwhelmed by excessive information and choices.
5. A frequent user needs a system that understands natural language input (e.g., "cheap dinner nearby") in order to avoid translating their needs into rigid filter settings.
6. A user browsing restaurant options needs reduced cognitive load in order to easily compare choices without remembering multiple details across listings.
7. A user needs transparent and trustworthy recommendations in order to feel confident that the suggested restaurants match their needs and are not biased (e.g., ads).
8. A user needs proactive decision support in order to move from browsing to choosing without unnecessary hesitation or repeated scrolling.